



Polyelectrolyte complex composite nanofiltration membranes fabricated in two steps with hydrocarbon-based sulfonated ionomer and polyethyleneimines

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ABSTRACT

This study presents a novel approach to fabricating chlorine-stable nanofiltration (NF) membranes using a scalable two-step dip-coating process using the hydrocarbon-based ionomer Nexar and branched polyethyleneimine (PEI) as polyelectrolyte. A thin-film composite (TFC) membrane was first formed by casting Nexar onto an aminated polyacrylonitrile (aPAN) support, followed by complexation with PEI to create a polyelectrolyte complex (PEC) film, to yield PEC-NF membranes. The influence of solvent systems, drying conditions and cast film thickness on the performance of the Nexar TFC membrane has been studied initially. With the best performing Nexar TFC prototype, it has been found that molecular weight (MW) of PEI has a profound influence on PEC-NF membrane structure and performance. A comprehensive characterization of the membrane structures has been performed by scanning electron microscopy, atomic force microscopy, TOF-SIMS analyses, as well as zeta potential and contact angle measurements. Compared to the TFC precursor, the PEC-NF membranes exhibited improved multivalent salt rejection as well as higher rejection of sucrose and organic micropollutants, particularly for neutral and hydrophobic compounds, at either reduced (low MW PEI) or slightly increased (high MW PEI) permeance. All membrane variants demonstrated excellent stability under prolonged exposure to active chlorine (500 ppm NaOCl, pH 9). Increasing feed salt concentration yielded markedly decreasing salt rejection and increasing solution permeance. Remarkably, the rejection of neutral solutes such as sucrose (~90 % for PEC-NF with low MW PEI) was not influenced by feed salt concentration in the range of 0.2–10 g/L. This work highlights the potential of ionomer-based PEC membranes as a robust and tunable platform for advanced NF applications.

1. Introduction

Water scarcity is poised to become one of the major challenges of the coming decades, even in regions traditionally considered water-rich. Advanced water treatment technologies are required to address not only salinity removal but also the separation of organic micropollutants (OMP), such as pesticides, pharmaceuticals, hormones, and other endocrine-disrupting compounds [1,2]. Nanofiltration (NF) has gained attention in water treatment due to its moderate solute rejection combined with higher permeability and lower transmembrane pressure requirements compared to reverse osmosis. While NF can effectively remove many highly charged and sufficiently large OMPs, its rejection

efficiency is highly solute-specific, and for some OMPs, incomplete rejection may lead to their accumulation in the treated effluent [3,4]. Most current NF membranes are thin-film composites (TFC), typically with some kind of polyamide (PA) layer formed through interfacial polymerization on a porous support membrane [5]. These membranes primarily separate solutes based on size exclusion, with molecular weight cut-offs (MWCO) typically in the range of 200–1000 Da. However, charge-based mechanisms such as Donnan exclusion as well as dielectric exclusion can play a secondary role in enhancing the rejection, depending on membrane surface charge and solute charge [6]. Despite their effectiveness, PA TFC membranes face critical drawbacks related to fouling and chemical vulnerability. In particular, their chemical stability

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remains a significant challenge because PA barrier films are susceptible to degradation by oxidants like hypochlorite ("active chlorine"), a commonly used disinfectant in water treatment processes [7].

A promising alternative to traditional polyamide NF membranes is the use of polyelectrolyte complexes (PEC) as the selective film [8]. PEC membranes can be fabricated using the layer-by-layer (LbL) assembly of oppositely charged polyelectrolytes on a porous support, offering tunable ion selectivity by varying polymer composition, the number of layers as well as deposition conditions, and resulting surface charge density [9]. Such multilayer polyelectrolyte membranes can be prepared to have a thin (≤ 100 nm) barrier film and have demonstrated good separation performance and stability in various applications. However, LbL assembly is a labor-intensive, multistep process, and the resulting structures, held together by reversible ionic bonds, may destabilize under high salt concentrations or extreme pH conditions. Indeed, recent studies have shown that highly saline environments can screen the electrostatic interactions in PEC films, leading to diminished separation performance or delamination from the support membrane [10]. Moreover, the mechanisms governing salt rejection and selectivity in PEC membranes are not yet fully understood, complicating predictive design and optimization [11].

To overcome the limitations of both PA TFC and conventional PEC multilayer membranes, recent research introduced a conceptually different approach for preparing PEC NF membranes involving a simple, two-step coating process using oppositely charged polymers [12]. In this method, an amphiphilic negatively charged ionomer is first deposited as a base coating to form a water-stable selective skin on a porous support membrane, followed by the application of a cationic polyelectrolyte to create a PEC TFC membrane. The proof-of-concept has been done with Nafion, a fluorinated sulfonic acid group containing ionomer that has originally been established as material for proton-exchange membranes [13]. By adjusting the polycation (e.g., polyvinylamine) concentration, resulting membrane charge and selectivity can be systematically controlled. This approach enables fine-tuning of separation performance characteristics without relying on labor-intensive LbL techniques. Nevertheless, concerns regarding the use of an expensive and unsustainable fluorinated ionomer and open questions related to precise control of molecular weight cut-off still remain.

The nanoscale morphology of ionomer films plays a pivotal role in determining their transport properties and separation performance. As reviewed by Li and Guiver, ion-conducting nanochannels formed by self-assembly in block or comb-shaped aromatic copolymers with ion-exchange group containing segments provide efficient pathways for water and hydrated ions, enabling high ionic conductivity and selective transport, that is critical for electrochemical applications but can also be relevant for water purification [14]. These nano-scale morphologies typically result from phase separation between hydrophilic ionic domains and hydrophobic matrix blocks, that enables tuning of the ion transport mechanism, either via interstitial water pathways or hopping between charged groups, thereby impacting flux, selectivity, and chemical resistance. In a study with sulfonated aromatic multiblock copolymers and the focus on water desalination, Park et al. emphasized the importance of polymer backbone chemistry and block copolymer architecture for achieving competitive desalination performance and chlorine tolerance of ionomer TFC membranes [15]. Wieczorek and Ulbricht demonstrated that the positively charged counterparts, i.e. aromatic multiblock copolymers with quaternary ammonium groups, can also be processed by film casting into nanochannel-in-matrix structures that yield TFC membranes with attractive intrinsic desalination performance and high chemical stability, including resistance to chlorine exposure [16]. In addition, Fischer et al. illustrated for cationic aromatic ionomers how the choice of casting solvent and post-treatment conditions (e.g., thermal or solvent annealing) can direct the formation of hydrophilic nanodomains in a hydrophobic matrix during membrane solidification, thereby offering another degree of control over membrane structure and function [17].

In this context, the sulfonated pentablock copolymer Nexar has garnered attention for its ability to form robust microphase-separated morphologies with high water uptake and tunable ion transport properties [18]. It has also been demonstrated that Nexar membranes and ionomers are competitive with their fluorinated Nafion counterparts as proton-conducting materials for hydrogen fuel cell applications [13]. As shown by Geise et al., variations in sulfonation degree and casting conditions significantly influence water and salt permeability in Nexar films, highlighting the critical link between charge density, morphology, and separation behavior [19]. These features, combined with Nexar's hydrocarbon-based backbone, render it a promising, scalable alternative to fluorinated ionomers like Nafion for the development of PEC-enabled NF membranes.

Building on these advances, the aim of this study was to develop novel PEC-enabled nanofiltration membranes that are suitable for tertiary wastewater treatment, with one focus on achieving high rejections for organic micropollutants while ensuring stability of separation performance. The commercial ionomer Nexar was used in this work. In one previous study, Nexar had been coated onto a positively charged porous support membrane to obtain a NF membrane that had then been studied in terms of heavy metal ion rejection [20]. In this work, anionic Nexar was combined with branched polyethyleneimine (PEI) as the polycation, and a scalable two-step dip-coating process was used to create PEC TFC membranes on an aminated polyacrylonitrile (aPAN) support membrane. Key variables such as solvent system, coating thickness, and post-treatments were examined to understand their impact on membrane morphology, structure, and separation performance. Detailed studies of the rejection of salt as well as sucrose and polyethyleneglycol (PEG) at varied conditions were performed to gain understanding regarding separation mechanisms. The rejection of various organic micropollutants was also studied. The membranes' resistance to active chlorine exposure was investigated to demonstrate the oxidation stability. Ultimately, this work contributes to bridging the gap between laboratory-scale PEC membrane innovations and scalable, robust membrane systems for real-world water reuse applications.

2. Materials and methods

2.1. Materials

Polyacrylonitrile (N-PAN; 80 kDa, 94 % acrylonitrile, 6 % methyl acrylate) was received from Dolan GmbH, Germany. The nonwoven Noxatexx 2471 PE/PP was obtained from Freudenberg SE, Germany. The ionomers NEXAR MD9200 (ion-exchange capacity, IEC = 2.0 meq/g) and NEXAR MD9100 (IEC = 1.0 meq/g) were donated by Kraton Corporation, USA. Branched polyethyleneimine (PEI) with molecular weights of approximately 600 Da and 25 kDa was purchased from Sigma-Aldrich. The chemical structure of the two components of the PEC membranes is shown in Fig. 1.

Ethylene diamine (≥ 99 % purity) was purchased from Sigma-Aldrich. Toluene (≥ 99.8 % purity), isopropanol (≥ 99.8 % purity), tetrahydrofuran (THF, ≥ 99.8 % purity), and sodium chloride (≥ 99.5 % purity) were obtained from Fisher Chemical. n-Propanol (99.5 % purity), calcium chloride (≥ 94 % purity) and polyethylene glycol (PEG 600) were supplied by Carl Roth. Sodium sulfate (99.2 % purity) was

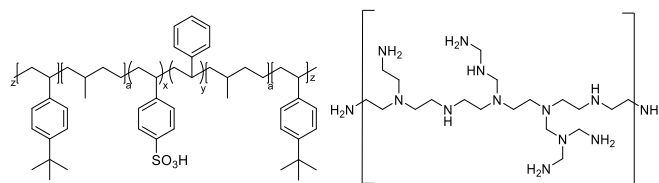


Fig. 1. Ionomer Nexar (left) and polyelectrolyte, i.e., branched polyethyleneimine (right).

purchased from VWR Chemicals. Magnesium sulfate (≥ 99 % purity), glucose (≥ 99 % purity) and sucrose (≥ 99 % purity) were obtained from AppliChem. Technical ethanol (96.4 % purity) was supplied by Caldic. Sodium hypochlorite solution (10–15 % active chlorine) was obtained from Thermo Scientific. Sodium hydroxide solution (0.1 M) and hydrochloric acid solution (0.1 M) were both purchased from Bernd Kraft. Sulfadiazine (>99.0 % purity), sulfamethoxazole (>98.0 % purity), sulfamethazine (>98.0 % purity), carbamazepine (>97.0 % purity), ibuprofen (>98.0 % purity), and triclosan (>98.0 % purity) were all obtained from TCI Chemicals. All chemicals were used as received without further purification unless stated otherwise. DI water was obtained by ion exchange, while ultrapure water was produced with a Millipore Milli-Q Academic® system.

2.2. PAN support membrane fabrication

The casting solution was prepared by gradually dissolving dry N-PAN into the appropriate solvent under continuous stirring. The recipe was 15 wt% N-PAN, 4 wt% LiCl, 7 wt% ethanol and 74 wt% DMSO, adapted from literature [21]. The mixture was then stored for at least 24 h at 80 °C in a convection oven, with intermittent manual stirring to ensure complete polymer dissolution. Membrane casting was carried out using a Philos pilot-scale casting machine (PHILOS, South Korea) at room temperature (20–22 °C) and moderate relative humidity (40–60 %). The nonwoven fabric served as the substrate, and the casting knife gap was set to 200 μm . The dope solution was applied at 7 cm/s, resulting in an air contact time of approximately 2.5 s before the film entered the coagulation bath containing water, positioned 20 cm downstream. After coagulation, the membrane was rinsed three times in DI water (containing 0.01 M NaN_3) for 30 min and then stored overnight in the same solution to stabilize the structure.

The amination reaction (see Fig. 2) was carried out as a one-pot process at 60 °C for 24 h in a 1 L 90:10 mixture of technical ethanol and DI water [22]. The reaction vessel was a 1 L Schott bottle. The membrane area was 600 cm^2 (30 $\times$ 20 cm), and the membrane sheet was rolled with the non-woven side facing the glass wall of the bottle. Ethylenediamine (EDA) was added in a tenfold molar excess relative to the number of repeating units of PAN (deduced from PAN mass per membrane area). The reaction mixture was stirred at 500 rpm. After completion, the obtained aPAN membrane was washed with ethanol and stored in DI water (with 0.01 M NaN_3). Before use as support for TFC membrane fabrication the membrane dried under ambient conditions after solvent exchange to ethanol.

The ultrafiltration-type membrane was tested in custom-built filtration cells with an effective membrane area of 4.1 cm^2 . Prior to characterization, membranes were compacted for 30 min at 0.5 bar, followed by three 5-min measurements of pure water permeance (PWP). Subsequently, the molecular weight cut-off (MWCO) was determined using dextran solutions at a flux ≤ 10 LMH. Further experimental details are described elsewhere [23].

2.3. TFC and PEC membrane fabrication

A Bungard RDC15 dip-coating system (withdrawal speed range: 30–1010 mm/min) was employed. The coating frame was constructed from stainless steel and allowed for an open membrane area of 240 mm

$\times$ 60 mm. It contained 28 holes designed to accommodate 28 polyamide screws for securing the membrane sheet in place.

For TFC membrane fabrication, dried Nexar was dissolved in the respective solvent (concentration provided in wt%) and stirred for at least 48 h prior to use. Three solvent systems were employed: a mixture of toluene and isopropanol (85:15 v/v) representing a non-polar solvent system with moderate vapor pressure, toluene and n-propanol (1:1 v/v) representing a somewhat more polar solvent system with a vapor pressure that is lower than that of first one, and tetrahydrofuran (THF), representing a moderately polar solvent with very high vapor pressure (see Supplementary Information, SI, section S1).

The dip-coating process for preparing TFC membranes involved a controlled sequence of steps to apply a uniform polymer film onto the porous support membrane. First, the dry support membrane was mounted in the stainless steel frame and vertically immersed into the Nexar-containing polymer solution. After complete submersion, the membrane was kept in the solution for a defined impregnation time of 10 s to ensure sufficient wetting. Subsequently, the membrane was withdrawn from the solution at a controlled speed ranging from 2 to 15 mm/s, allowing a thin, uniform coating layer to form on the membrane surface. Following the dip-coating process, the frame with the coated membrane was removed from the dip-coater, placed horizontally (membrane surface upwards) and covered with aluminum foil, leaving a gap of approximate 0.5 cm height, thereby creating a gas space of approximately 72 cm^3 relative to the 144 cm^2 membrane area for “slow” evaporation. “Fast” evaporation was performed without covering the membrane.

For PEC formation, dry TFC membranes within the sealed frame, exposing only the top surface, were placed into a 2 L water bath containing the polyelectrolyte solution at room temperature. The polyelectrolyte concentration was typically 0.01 wt%, and membranes were typically kept in the solution for 30 s. Subsequently, the membranes were thoroughly rinsed and then washed in DI water for 1 h and kept in 0.01 M NaN_3 DI water.

For annealing in solvent vapor (THF), the PEC membrane was placed inside a desiccator (volume: 25 L) along with a 200 mL bowl of liquid THF. The theoretically required volume of liquid THF to saturate the gas phase in the desiccator (V_s) was estimated using the ideal gas law:

$$V_s = \frac{M_s \cdot PV}{\rho_s \cdot RT}$$

For thermal annealing in liquid water, the membrane was immersed in a 2 L water bath, which was heated to 60 °C in an oven for 7 days.

2.4. Membrane structure characterization

Surface morphology was examined using scanning electron microscopy (SEM; instrument ESEM Quanta 400 FEG, FEI, USA). For cross-sectional analysis, the samples were cooled with liquid nitrogen and fractured. Subsequently, they were coated with an ~ 3 nm thin Au/Pt layer (80:20) using a K550 Sputter Coater (Emitech, Ashford, UK).

To investigate the nanoscale morphology, atomic force microscopy (AFM) analyses were performed on all relevant membranes in wet state. Measurements were carried out using a Bruker Dimension FastScan® system in PeakForce Tapping® mode to obtain topographical images. Before the wet scans, samples were soaked for 3 days in DI water.

Time-of-flight secondary ion mass spectroscopy (TOF-SIMS) measurements were performed using a TOF.SIMS 5 instrument (IONTOF) equipped with a bismuth cluster ion source (Bi_3^+) operated in spectrometry and random scanning modes at 15 kV. The analysis area was set to 100 $\times$ 100 μm^2 with a resolution of 128 $\times$ 128 pixels. Negatively charged secondary ions were analyzed under negative polarity settings, and dead time correction was applied throughout all measurements. For selected mass peaks such as H^- and CN^- , the extended dynamic range mode was activated to prevent detector saturation. Depth profiling was carried out using non-interlaced sputtering with Ar^+ ions at 5 kV over a

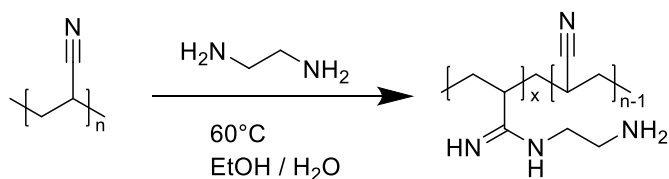


Fig. 2. Polymer-analogous amination reaction of PAN with EDA to yield aPAN membrane.

400 × 400 μm² area, with analysis-to-sputter frame ratios of 1:4 and 1:1.

For XPS analysis, membrane samples were first dried. The analysis was conducted on the top surface of the membranes using a PHI VersaProbe II with an Al Kα X-ray source, resulting in a photon energy of 1486.6 eV. The etching of the membrane samples for depth profiling was done using a 1 keV Ar⁺ ion beam. The estimated etching rate was 1–2 nm/min.

The surface charge of the membranes was analyzed using the electrokinetic analyzer SurPASS (Anton-Paar GmbH, Austria). The gap height between two identical membrane samples forming the flow channel was set at 100 (±10) μm. A total of 550 mL of an aqueous solution of 1 mM KCl was employed as an electrolyte with its pH value pre-adjusted to approximately 3. Measurements were performed in the range from pH ~3 to pH ~10, with each new pH value adjusted by adding a KOH solution. Rinsing time was 300 s for each step, ensuring accurate and stable pH values at the membrane surface. Streaming current was measured as a function of pressure difference (which is a function of flow rate). The data were automatically processed by the SurPASS software according to the Helmholtz-Smoluchowski equation to yield the zeta potential as function of pH value.

For water contact angle measurement in the sessile drop mode, dry membranes were pasted on glass slides to obtain flat samples, and then analyzed using the contact angle system OCA15 (Dataphysics) at 25 °C. Drops of water (5 μL) were placed on at least five different positions on every sample, and average values along with standard deviations were reported.

2.5. Filtration setups, test conditions and determination of flux/permeance

Membrane performance was evaluated using two complementary nanofiltration (NF) systems: a small-scale dead-end NF system and a cross-flow NF unit.

The dead-end filtration system was equipped with a magnetic stirrer to minimize concentration polarization. The dead-end cell had an effective membrane area of 9.62 cm² and a total cell volume of 100 mL. Filtration experiments were conducted at a constant applied pressure of 15 bar. Prior to first pure water permeance (PWP) measurement, each membrane was pre-compacted for 30 min to ensure stabilization of flux. Prior to each subsequent filtration, membranes were thoroughly rinsed with the feed. Then 100 mL of feed were filled into the cell, filtration was started and the first ~2 mL of permeate was discarded to eliminate dead volume effects. Filtration proceeded thereafter until a total permeate volume of ~2 mL was collected.

Complementary testing was performed using a cross-flow unit (CUBE 80-2 VA, SIMA-tec GmbH, Germany) with an effective membrane area of 84 cm² and a feed volume of 2 L. This setup enabled dynamic testing under continuous flow conditions, providing insights into solute rejection and flux performance under more realistic operating scenarios including long-term evaluation. Applied pressure was 15 bar if not specified otherwise. The first measurement, conducted after a 30-min compaction phase, was determination of PWP, with data recorded over a 15-min period. Subsequently, the feed tank was emptied, a salt solution was introduced and the membrane was allowed to reach a steady state for at least 30 min until a stable conductivity value in the permeate was observed. A similar procedure was applied with organic solutes, with reaching steady state monitored using total organic carbon (TOC) analysis. Thereafter, the permeate flux was measured for an additional period of 10 min. Between each solute test, the membrane system was thoroughly rinsed with ultrapure water until the conductivity or TOC values in the permeate approached baseline (pure water) levels. Subsequently, PWP was measured again for at least 10 min to assess membrane performance recovery and stability.

Filtration tests were conducted using aqueous model solutions containing either inorganic salts (1 g/L) or organic solutes (0.2 g/L), depending on the specific test.

Pure water permeance (PWP) was calculated as follows:

$$\text{PWP} = \frac{V}{A \cdot t \cdot \Delta P}$$

With V as the volume of the permeate, A as the effective membrane area, t as the permeation time and ΔP as the applied transmembrane pressure.

Salt solution permeance (SSP) was calculated considering the osmotic pressure of the feed:

$$\text{SSP} = \frac{V}{A \cdot t \cdot (\Delta P - \Pi)}$$

With the osmotic pressure (Π):

$$\Pi = i \cdot c \cdot R \cdot T$$

With i as the van 't Hoff factor, c as the molar concentration of the solute, R as the universal gas constant and T as the temperature.

The rejection of selected organic micropollutants (OMP) was investigated using cross-flow filtration only. The following substances were selected as model OMP: sulfadiazine (1), sulfamethoxazole (2), sulfamethazine (3), carbamazepine (4), ibuprofen (5), and triclosan (6). For the preparation of feed solutions, 50 mg of each OMP was first dissolved in 10 mL of ethanol and subsequently diluted into 5 L of ultrapure water. Two sets of mixed solutions were prepared: one containing OMP 1–4 ("Mix 1"), and another containing OMP 5&6 ("Mix 2"). In addition to testing a mixture of OMP, individual experiments were performed for each compound as well. For these single-solute tests, time-resolved measurements were carried out to monitor rejection behavior and assess potential adsorption onto the membrane or system components. The protocols for the NF experiments with the mixtures were selected so that membranes had reached a steady state, i.e., removal by adsorption has no influence on permeate concentration. OMP concentrations in feed and permeate samples were quantified using high-performance liquid chromatography (HPLC).

2.6. Analysis of solute concentrations and determination of solute rejection

Electrical conductivity of the salt solutions was measured using a Schott Instruments Lab 960 conductivity meter. Measurements were conducted at room temperature (~25 °C). For each salt calibrations, dependence of conductivity on salt concentration, were also performed and used for estimation of the respective feed and permeate concentrations.

Total organic carbon (TOC) was analyzed using a TOC-V CPN analyzer (Shimadzu) equipped with an ASI-V autosampler. Inorganic carbon (IC) was removed by acidification and sparging prior to oxidation. TOC was calculated as the difference between total carbon (TC) and IC (TOC = TC – IC). Calibration was carried out using potassium hydrogen phthalate standards.

The analysis of OMP was performed using HPLC on a Shimadzu Nexera Series 40 system, consisting of an SCL-40 system controller, DGU-403 degassing unit, LC-40B XR solvent delivery module, SIL-40c XR autosampler, RF-20A fluorescence detector, and an SPD-M30A photodiode array detector. Separation was carried out using an EC 100/4 Nucleosil 120-3 C18 column. The mobile phase consisted of ultrapure water with 0.01 M formic acid and acetonitrile (ACN). Isocratic elution was applied (no gradient), with a flow rate of 0.8 mL/min and a total run time of 15 min per measurement. The injection volume was 10 μL, and the column temperature was maintained at a constant 35 °C. Detection was performed using the photodiode array detector at a wavelength of 260 nm. Different ACN concentrations in the mobile phase were used depending on the OMP mixture. OMPs 1–3 (Mix 1) were analyzed using a method with 15 % ACN. OMP 4 (also part of Mix 1) and OMP 6 (from Mix 2) were analyzed using a separate method with 60 % ACN. OMP 5 (from Mix 2) required a third method using 20 %

ACN.

The obtained concentrations from the conductivity, TOC and HPLC measurements were used to calculate the rejection R by following equation:

$$R = 1 - \left(\frac{C_p}{C_F} \right) * 100\%$$

With C_p as the concentration in the permeate and C_F as the concentration in the feed.

2.7. Reproducibility and error of flux and rejection data

For all membrane types, at least two coupons obtained from one membrane sheet were tested. In the cases where only such two measurements had been performed, the data for permeance, MWCO or solute rejection are reported as average value without error range/bars. The difference between the two values was less than 5 %. In all other cases, at least 3 coupons from one membrane sheet or at least 4 coupons from two membrane sheets were tested. The respective data for permeance, MWCO or solute rejection are reported as average value along with error calculated as the standard deviation from the average value.

3. Results and discussion

This section is organized thematically to follow the membrane

development process. It first covers fabrication and post-modification of the PAN support membranes. It then details formation and optimization of the Nexar barrier film to yield TFC membranes. NF separation performance is thereafter used to assess the effects of converting Nexar into PEC membranes. Then, the structural characterization of the membranes is presented. Finally, the NF performance under varied feed and operational conditions and the capability regarding rejection of organic micropollutants is analyzed. The PEC TFC membrane fabrication process is illustrated in Fig. 3.

3.1. PAN-based support membrane

To establish a suitable porous support membrane for TFC fabrication, a non-woven supported PAN membrane was fabricated via film casting and liquid non-solvent induced phase separation (NIPS). With the aim to obtain a membrane that has a MWCO (<70 kDa) that is small enough to prevent penetration of Nexar (MW ~ 112 kDa) into the pores and a high PWP, conditions reported in literature [21] had been used as starting point for lab-scale experiments to establish suited casting conditions that were then used on the pilot-scale machine (see section 2.2). Data are presented in Table 1.

This base PAN membrane was sequentially modified by amination and solvent treatment. First, the PAN membrane was reacted with ethylenediamine (EDA) in an ethanol/water mixture at 60 °C for 24 h to introduce amine functionalities, producing the aminated PAN (aPAN) membrane (cf. Fig. 2). Subsequently, the aPAN membrane was subjected

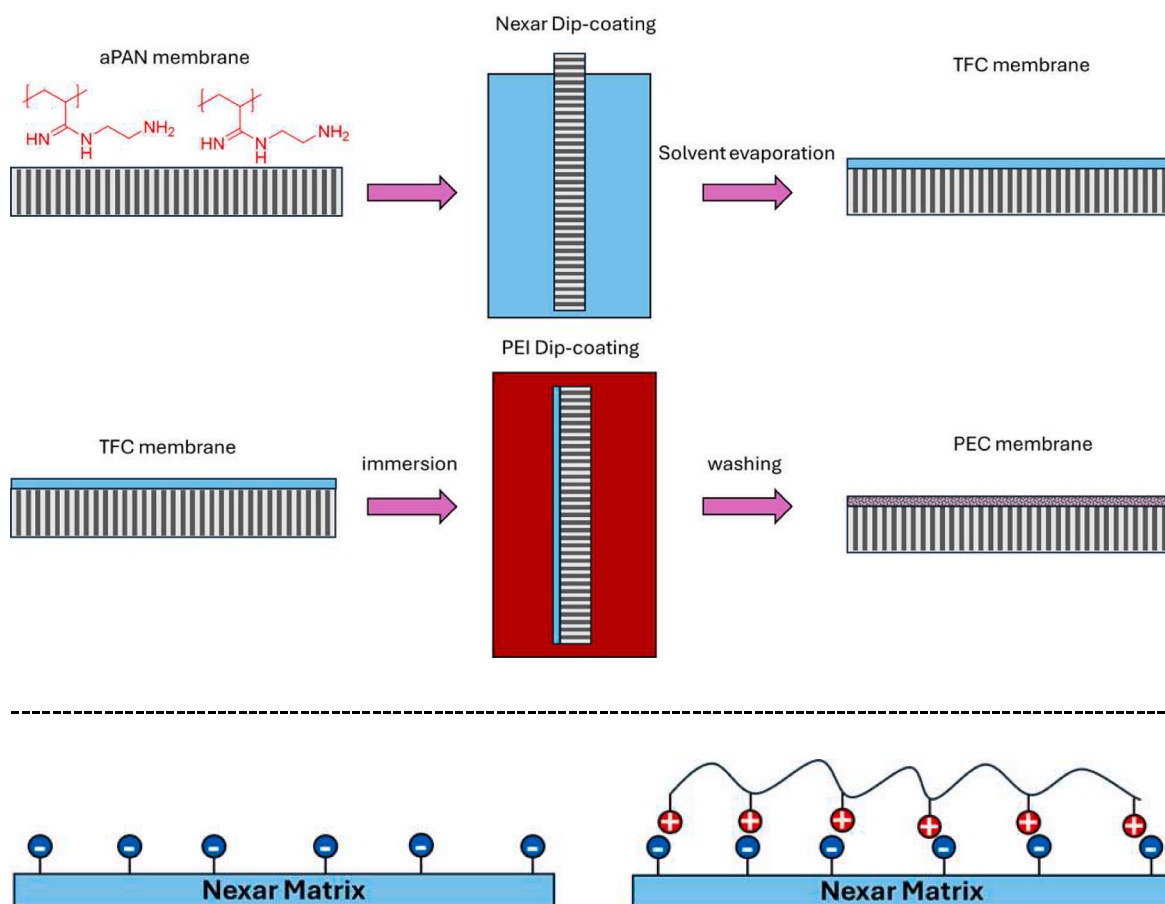


Fig. 3. Top. Schematic illustration of the fabrication process for thin-film composite and polyelectrolyte complex membranes. First, the separately prepared aminated polyacrylonitrile (aPAN) support membrane is dip-coated with a Nexar solution, followed by solvent evaporation to yield the Nexar TFC membrane. Second, the TFC membrane is immersed in an aqueous polyethyleneimine (PEI) bath (typically 0.01 wt%, 30 s) to form the PEC membrane. Bottom. Schematic illustration of the PEC formation via ionic interaction between Nexar's sulfonic acid groups and PEI's protonated amino groups. These interactions can take place on the outer surface of the Nexar film as well as in the ionic domains (channels/pores) in the Nexar matrix. The complexation can lead to charge-balanced domains as shown in the cartoon, but other stoichiometries are also possible [12].

Table 1

Pure water permeance (PWP) and molecular weight cut-off (MWCO) of as fabricated and modified PAN membranes.

Membrane	PWP [LMH/ bar]	MWCO [kDa]
PAN	611 ± 32	63.2 ± 9.6
aPAN	313 ± 17	30.8 ± 7.3
aPAN (after exposure to and drying from toluene: isopropanol (85:15 v%))	156 ± 11	24.0 ± 5.6

to a simulated dip-coating process, involving drying from ethanol, then soaking in a toluene:isopropanol (85:15 v%) mixture and drying, mimicking conditions during the Nexar film formation (cf. Fig. 3).

As shown in Table 1, these treatments caused a progressive reduction in PWP and MWCO. After amination, PWP dropped from 611 ± 32 to 313 ± 17 LMH/bar, and MWCO dropped from 63.2 ± 9.6 to 30.8 ± 7.3 kDa. The subsequent solvent treatment and drying further reduced PWP to 156 ± 11 LMH/bar and MWCO to 24 ± 5.6 kDa. This decline is attributed to thermally induced annealing during the amination step and to pore constriction by solvent exposure. The pore and membrane structure observed by SEM can be found in the SI (section S2). Elevated temperature likely caused densification of the pore structure, while solvent interactions contributed to further structural tightening. Despite this, the final aPAN support membrane retained a PWP that is an order of magnitude higher than that of typical NF membranes, confirming its suitability as a support for high-performance TFC membranes [24,25]. And the MWCO value is certainly low enough to prevent Nexar penetration and blocking of support pores.

3.2. Influence of dip-coating conditions on Nexar TFC membranes

To optimize the formation of the selective Nexar film in TFC membranes, the effects of dip-coating parameters, including the solvent system, withdrawal speed and evaporation rate were systematically investigated. These parameters were varied to study how they influence membrane morphology and, consequently, separation performance. The aim was to identify processing conditions that promote the formation of a uniform, defect-free selective film enabling high water permeance and effective solute rejection.

3.2.1. Effects of solvent choice and evaporation rate

To evaluate how solvents affect the formation and performance of the Nexar film, membranes were prepared using three solvent systems: toluene:isopropanol (85:15 v%), toluene:n-propanol (50:50 v%), and pure tetrahydrofuran (THF), all with 3 wt% polymer concentration and processed under the same dip-coating and slow drying conditions. These solvents differ in polarity and Hansen solubility parameters, as well as volatility (see section S1 in SI), which influences how the block copolymer segments self-assemble during film formation. Nexar, being an amphiphilic block copolymer with sulfonated hydrophilic blocks and hydrophobic blocks, can microphase-separate into different morphologies depending on solvent quality for each block [18,26–30].

The results show substantial differences in separation performance (Table 2). The toluene:isopropanol system yielded a balanced membrane with moderate water permeance (2.2 LMH/bar), good sucrose rejection (~55 %) and the highest NaCl rejection (78 %). The membrane cast from n-propanol:toluene showed a much higher PWP (~3.9 LMH/bar) but very poor rejection for sucrose (~20 %) and NaCl (~5 %). The THF-cast membrane had low PWP (~1.1 LMH/bar) and good sucrose rejection (~65 %), but low rejection for NaCl (35 %). These observations indicate that solvent polarity and evaporation behavior strongly influence the final membrane structure. The toluene:n-propanol system, that is not very volatile, likely created a well solvating environment that favored the formation and expansion of ionic domains in a slowly solidifying hydrophobic matrix, leading to hydrophilic pathways that

Table 2

Influence of the solvent system and of the evaporation rate for toluene:isopropanol solvent mixture on the separation performance of the TFC membranes (0.2 g/L sucrose, 1 g/L NaCl; 15 bar applied pressure).

Solvent (evaporation)	PWP [LMH/ bar]	Sucrose rejection [%]	NaCl rejection [%]
toluene:isopropanol 2 h (fast ^a)	1.2	30	65
toluene:isopropanol 12 h (slow ^b)	2.2 ± 0.3	54.6 ± 4.6	77.6 ± 6.7
toluene:n-propanol (slow ^b)	3.9	20	5
THF (slow ^b)	1.1	65	35

^a Free evaporation.

^b Membrane covered to yield a saturated gas atmosphere.

allow sucrose (~342 Da; size ≤1 nm) to pass almost without hindrance and that do not hinder passage of hydrated Na⁺ and Cl⁻ ions. In contrast, THF, a fast-evaporating, good solvent for both blocks, may have promoted phase separation to yield size-selective pores (higher rejection for sucrose than for NaCl), but porosity/pore-connectivity or accumulation of charged groups in the pores were likely not well-developed. The balanced separation performance of the membrane from the toluene:isopropanol system suggests that this allows the best self-assembly toward a channel-in-matrix morphology with size-sieving properties and charged groups in the pores (higher rejection for NaCl than for sucrose).

To evaluate the impact of evaporation rate on membrane structure and performance, Nexar films were dip-coated from a toluene:isopropanol (85:15 v%) solution and then subjected to two drying conditions: covered with aluminum foil ("slow"; duration ~12 h) and uncovered ("fast"; duration ~2 h). All other parameters, including polymer concentration and withdrawal speed, were kept constant. Remarkably, all performance indicators deteriorated largely after fast drying, PWP declined from 2.2 to 1.2 LMH/bar, and the rejection values for sucrose and NaCl declined from 55 to 30 %, and from 78 to 65 %, respectively. This performance difference is attributed to a significantly different film formation process. Slow evaporation provides sufficient time for the sulfonated segments of the Nexar copolymer to self-organize into a percolating network of hydrophilic domains embedded in a co-continuous hydrophobic matrix. This channel-in-matrix structure facilitates water transport while maintaining size exclusion for solutes such as sucrose (~342 Da) and enabling a significant additional effect of charged groups in the pores on salt rejection (this will be investigated in more detail in subsequent sections). In contrast, rapid solvent evaporation likely leads to kinetic trapping of polymer in a less ordered morphology, limiting domain connectivity and potentially creating non-selective pathways. As a result, the fast-dried membrane has lower solute rejection and lower permeance. These findings are consistent with results of prior studies on ionomer thin films, where slower solvent evaporation has been shown to reduce defect density and enhance ionic domain connectivity [31,32]. Exactly the same trends, major increases of PWP and solute rejection, have also been observed for TFC membranes with positively charged ionomer films after the cast films were dried at lower temperature, i.e., slower rate [16]. Notably, for such self-assembling ionomer systems, the improved permeance does not necessarily come at the cost of solute rejection; slow solidification allows to achieve a much more favorable balance of permeability and selectivity.

Given the comparatively poor performance of both, toluene:n-propanol (low rejection) and pure THF (low permeance), these solvent systems were not pursued further. Instead, the majority of subsequent experiments focused on the toluene-rich solvent system (toluene:isopropanol 85:15 v%), which allows to produce a well-balanced membrane.

3.2.2. Effect of dip coating speed

Besides solvent effects, the thickness of the Nexar coating (controlled by the viscosity and surface tension of the solution and the dip-coating withdrawal speed) is another important parameter. In dip-coating, higher withdrawal speeds generally entrain a thicker liquid film, leading to a thicker dried film, whereas lower speeds yield thinner films. However, very low speeds can lead to non-uniform films due to gravitational drainage and instabilities [33]. A series of membranes was prepared using the toluene:isopropanol solvent with varying withdrawal speeds (i.e., 120, 520, and 820 mm/min). The measured dry film thickness (obtained from cross-sectional SEM images of the TFC membranes; see section S2 in SI) and resulting PWP and solute rejections are given in Table 3. After using 120 mm/min, the Nexar film was $\sim 0.40 \mu\text{m}$ thick; for 820 mm/min, thickness increased to $\sim 1.5 \mu\text{m}$ (nearly $3 \times$ thicker). Intuitively, one would expect the thinnest film to yield the TFC membrane with highest water permeance (since it presents the lowest transport resistance), but possibly the lowest rejection if defects are present. Surprisingly, the obtained results did not show a simple inverse relationship between film thickness and flux. The thickest film ($\sim 1.5 \mu\text{m}$, for 820 mm/min) actually yielded the highest permeance (2.2 LMH/bar) followed by the medium thickness ($\sim 0.8 \mu\text{m}$, for 520 mm/min: 1.3 LMH/bar) and the thinnest film ($0.4 \mu\text{m}$, for 120 mm/min: 0.5 LMH/bar). Furthermore, the thickest film also achieved the highest sucrose rejection ($\sim 54 \%$), whereas the thinnest film yielded only $\sim 22 \%$ sucrose rejection. The NaCl rejection trend (820 mm/min: $\sim 79 \%$, 520 mm/min: $\sim 57 \%$, 120 mm/min: $\sim 65 \%$) is somewhat non-monotonic, but generally the highest salt rejection was also observed for the thickest film. These trends suggest that film quality and microstructure are more important than minimizing thickness. It appears that the 120 mm/min coating (very thin) may have suffered from support membrane pore blockage as well as incomplete support surface coverage or microscopic defects, overall drastically reducing its performance. The moderate speed (520 mm/min, $\sim 0.8 \mu\text{m}$) produced a thicker but perhaps still imperfect film. The highest speed (820 mm/min) yielded a robust, thick film that paradoxically yielded the best of both: highest flux and highest solute rejection. It appears that the high speed helped achieve a superior film because viscous drag dominates and facilitates the creation of a smooth film. At a thickness of $\sim 1.5 \mu\text{m}$, the Nexar film is likely to form a continuous, extrinsic defect-free network of ionic domains, which facilitates water transport via connected hydrophilic channels while still maintaining a relatively tight pore size for solute rejection [20]. Thicker films might also tolerate minor intrinsic defects without compromising overall rejection, as defects would be buried under more material. This finding highlights that simply minimizing film thickness does not guarantee better separation performance; instead ensuring the structural integrity of the selective film is critical. Similar observations were made by Zelner et al., who noted that an optimal polyelectrolyte complex film thickness, obtained by coating the ionomer Nafion in the first step, existed where flux was maximized without sacrificing rejection [12].

3.2.3. Prototype TFC membrane

The TFC membrane fabricated using a toluene:isopropanol (85:15 v %) coating solution serves as the performance benchmark in this study.

Table 3

Influence of dip-coating withdrawal speed for Nexar solutions (3 wt%) in toluene:isopropanol on pure water permeance and solute rejection of Nexar TFC membranes (0.2 g/L sucrose, 1 g/L NaCl; 15 bar applied pressure).

Dip-coating speed [mm/min]	Film thickness [μm]	PWP [LMH/bar]	NaCl rejection [%]	Sucrose rejection [%]
120	0.40 ± 0.05	0.5	65	22
520	0.78 ± 0.15	1.3	57	45.3
820	1.45 ± 0.10	2.2 ± 0.3	79.3 ± 7.3	54.6 ± 4.6

In this solvent system, the high fraction of less polar and less volatile toluene dominates the self-assembly process, allowing the slow organization of the hydrophobic matrix. As a result, the ionomer is expected to have formed a morphology consisting of disordered spherical sulfonated domains embedded within a continuous matrix of the hydrophobic blocks, as reported in the literature [18,26–30].

Fig. 4 provides an overview of nanofiltration performance for the Nexar TFC membrane characterized in the dead-end filtration system. The salt rejection pattern, i.e. high rejection of Na_2SO_4 and low rejection for CaCl_2 is characteristic of a negatively charged NF membrane, where Donnan exclusion preferentially blocks multivalent co-ions. The rejection of PEG 600 is 99 %. The estimated MWCO of $\sim 450\text{--}500 \text{ g/mol}$ is much lower than that of the base aPAN support ($\sim 24,000 \text{ g/mol}$ after exposure to the solvent and drying conditions used for dip-coating; cf. Table 1), confirming that the selective barrier in the TFC membrane is the Nexar film.

Zeta potential measurements of the TFC membrane, consisting of a Nexar film on an aminated polyacrylonitrile (aPAN) support, provide insights into the surface charge across a range of pH values (see section S3 in SI). The aPAN support, i.e. PAN modified with ethylenediamine, exhibits a positive zeta potential in the pH range up to ~ 6.5 , and increasingly negative values are observed as the pH further increases. The amino groups introduced during modification are protonated at lower pH values, resulting in a higher isoelectric point (point of zero charge) compared to PAN. In contrast, the Nexar TFC membrane maintains a negative zeta potential across all pH values. The presence of sulfonic acid groups in Nexar significantly influences the surface charge of the membrane. Importantly, the moderate positive charge of the aPAN support contributes to improved adhesion with the coated negatively charged Nexar film. In preliminary experiments (data not shown) using unmodified PAN membranes, delamination of the Nexar film was observed, i.e., TFC membranes were not stable enough.

3.3. Post-treatment of Nexar TFC membrane

While the prototype Nexar TFC membrane showed good rejection performance, its water permeance ($\sim 2.2 \text{ LMH/bar}$) was relatively low with regard to practical applications. Solvent or thermal annealing as a post-fabrication treatment to enhance the membrane permeability without sacrificing rejection was therefore explored. Solvent vapor annealing is known to induce morphological rearrangements in block copolymer films by allowing the polymer to partially re-solvate and re-organize. In particular, THF vapor can swell both the hydrophilic and

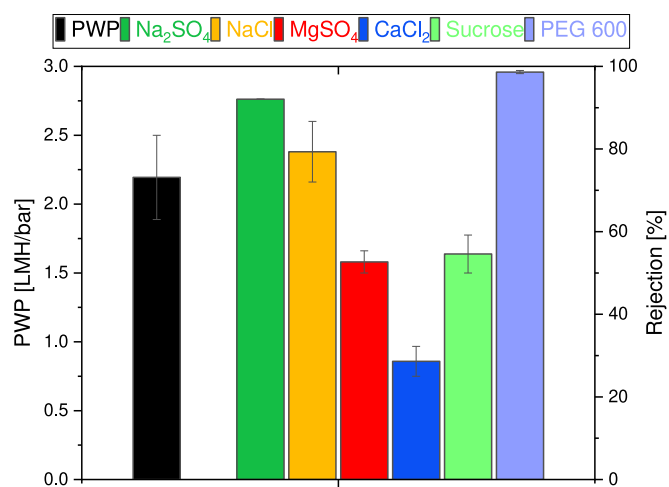


Fig. 4. Nanofiltration performance in dead-end filtration system (0.2 g/L organic solute, 1 g/L salt; 15 bar applied pressure) for Nexar TFC membrane (fabrication: 3 wt% Nexar in toluene:isopropanol (85:15 v%); dip-coating speed: 820 mm/min; slow evaporation).

hydrophobic blocks of Nexar (THF is a relatively neutral solvent for the polymer, see section S1 in SI), potentially driving the microstructure towards a more equilibrated state such as a lamellar or bicontinuous morphology [30]. After exposure of Nexar TFC membranes to THF vapor notable changes were indeed observed. The pure water permeance roughly doubled (from ~ 2.2 to ~ 4 LMH/bar). However, this came at the cost of reduced rejection for neutral organic solutes; sucrose rejection dropped from ~ 50 % to ~ 20 % (see section S4 in SI). This indicates that the transport pathways widened and therefore became less size-selective as a result of this annealing. It is inferred that the Nexar morphology may have transitioned towards a more ordered but also more porous structure, likely a lamellar structure with larger continuous channel domains. Prior work supports this interpretation: Mineart et al. showed that solvent annealing a sulfonated block copolymer could switch its morphology from cylindrical to lamellar, increasing water uptake [30]. In case of this study, such a lamellar structure might form larger straight channels through the film, boosting flux but also allowing more sucrose passage (hence lower rejection). This flux-selectivity trade-off highlights the challenge in membrane optimization: modifications that enhance flux (via increasing pore size or connectivity) often also increase the likelihood of solute passage.

Thermal annealing at 60°C in water for 7 days resulted in a significant improvement in solute rejection across all tested substances. For instance, sucrose rejection increased from approximately 50 % to 60 %, while Na_2SO_4 rejection surpassed 90 %. This slight enhancement is attributed to the formation of a denser membrane structure, likely caused by polymer domain reorganization during the hot water treatment. The improved rejection, particularly for neutral organic solutes and multivalent salts, suggests a reduced MWCO. However, this came at the cost of reduced membrane permeance, which decreased to 1.5 LMH/

bar (see section S4 in SI), due to the tighter morphology.

Other alternative treatments, including annealing in various other solvent vapors (e.g., ethanol, water) and at different temperatures in liquid water, were also explored. However, none resulted in membranes with performance clearly superior to the prototype Nexar TFC membrane. In general, when some treatments led to modest improvements in permeance, they caused a reduction in salt or neutral solute rejection. Therefore, the TFC membrane yielding the performance shown in Fig. 4 was selected as precursor for PEC formation and reference for comparison with PEC membranes.

3.4. Polyelectrolyte complex membranes – formation, separation properties and stability

With the Nexar TFC membrane established as the reference and precursor (cf. Section 3.2), the effect of contacting with polyethyleneimine (PEI) to form a polyelectrolyte complex (PEC) film was evaluated. Two molecular weights of PEI, 600 Da and 25 kDa, were used, resulting in two membrane variants (PEI-600 and PEI-25k). PEC formation was carried out by immersing the TFC membranes in a 0.01 wt% aqueous solution of the respective PEI for 30 s. Longer exposure time or higher PEI concentration lead to low pure water permeance (typically <1 LMH/bar; data not shown), a performance that is not competitive. These PEC-modified membranes, along with the unmodified TFC precursor, were then tested in a cross-flow setup to assess their performance. For each membrane type, two separate samples (from different coating batches) were evaluated to ensure reproducibility; Fig. 5 presents the average values with error bars representing the range of measured values. It was noted that in cross-flow, solute rejections were slightly higher, by a few percent, than in dead-end (cf. Fig. 4), due

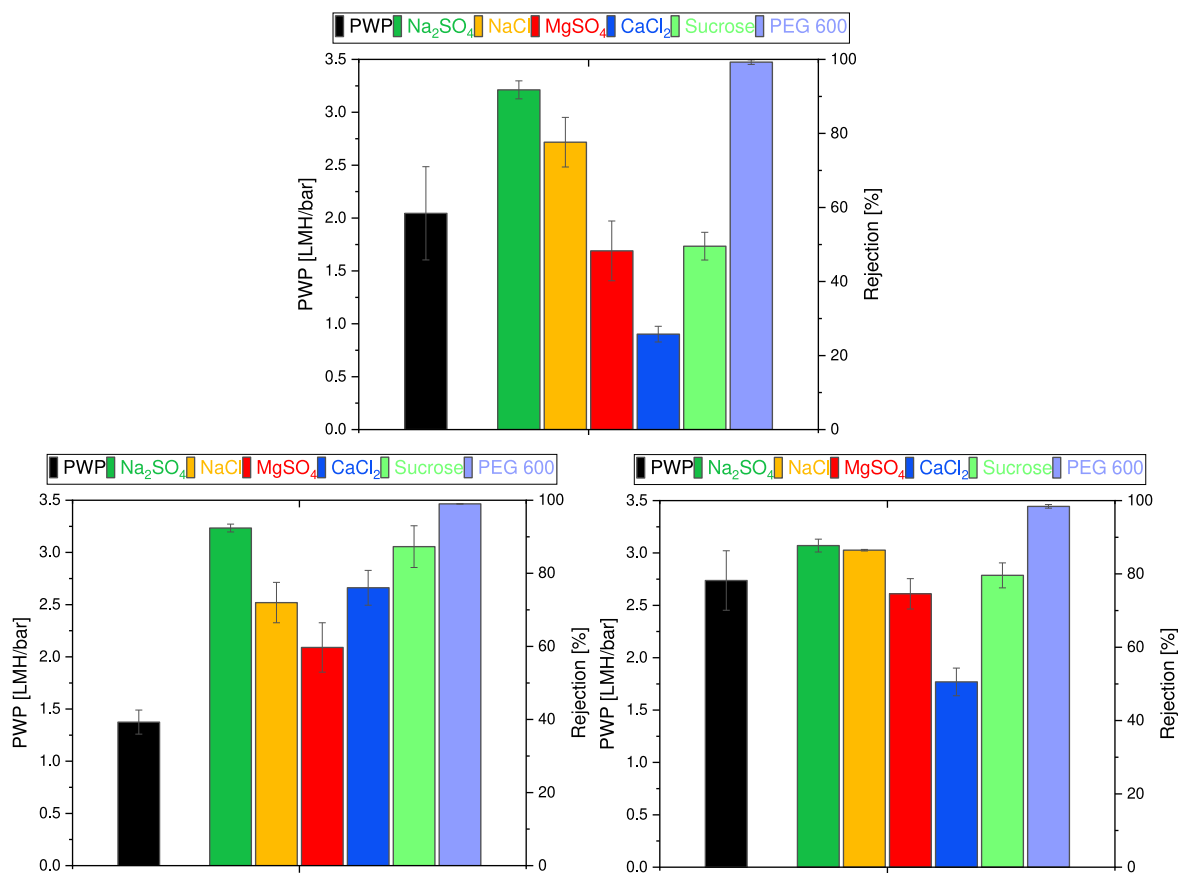


Fig. 5. Separation performance in cross-flow filtration (0.2 g/L organic solute, 1 g/L salt; 15 bar applied pressure). Top: Nexar TFC; Bottom-left: PEC PEI-600; Bottom-right: PEC PEI-25k. Nexar TFC membrane fabricated with 3 wt% Nexar in toluene:isopropanol (85:15 v%); dip-coating speed: 820 mm/min; slow evaporation. Errors indicating the range of measured values for two different separately fabricated sheets.

to the higher shear at the membrane surface reducing concentration polarization. Importantly, the overall trends remained consistent between the two modes.

Several clear effects of the additional PEI coating were observed. First, the addition of PEI (either molecular weight) caused a decrease in the MWCO of the membrane, indicating a tighter barrier pore structure. The estimated MWCO values (based on the rejection of neutral solutes like polyethylene glycols and sucrose) were: ~ 450 Da for TFC, ~ 350 Da for PEI-600, and ~ 400 Da for PEI-25k. Second, at a fixed feed concentration of 1 g/L, the PEI-coated membranes showed improved salt rejection, particularly for multivalent salts. For example, the Nexar TFC membrane rejected ~ 50 % of MgSO_4 , whereas PEC PEI-600 and PEC PEI-25k membranes rejected 60 % and 80 % of MgSO_4 , respectively. CaCl_2 rejection likewise increased (from ~ 25 % for TFC to ~ 75 % and 50 % for the two PEC membranes). This enhancement is phenomenologically attributed to the introduction of positively charged PEI, which moderates the influence of the negatively charged Nexar. By depositing PEI onto the sulfonated Nexar surface, a mixed-charge structure is created, potentially shifting the membrane surface toward neutrality or even a net-positive charge. This mixed-charge character favors the exclusion of cations, particularly multivalent ones, via Donnan exclusion. Importantly, what is referred to here as „charge density“ includes contributions from both fixed ionic groups (sulfonic acid in Nexar, protonated amine in PEI) and mobile counter-ions present within the hydrated channels in the membrane matrix. Zeta potential measurements (discussed in Section 3.5) support this interpretation, showing that PEI-25k in particular reduces the negative surface charge relative to the unmodified TFC. Consequently, divalent cations such as Ca^{2+} and Mg^{2+} experience weaker electrostatic attraction in the PEC-coated membranes, enhancing their rejection [34]. Notably, NaCl rejection did not improve as much with PEI coating, in fact at 1 g/L NaCl, TFC had ~ 70 % rejection, while PEC membranes had around 70–80 %.

In addition to Donnan effects, the improved rejection of multivalent salts can be further explained by combined contributions of steric and dielectric exclusion mechanisms [6]. Steric hindrance arises from the reduced effective pore size caused by PEI deposition, as revealed by higher sucrose rejection. Dielectric exclusion, which becomes significant when the membrane pores are small and the polymer matrix has a low dielectric constant compared to water, results from the energetic penalty associated with partial ion dehydration upon entry into a low-permittivity environment. Unlike Donnan exclusion, dielectric exclusion affects both cations and anions equally, more strongly the multivalent ions, and is independent of the membrane's net charge. The major increase in rejection of multivalent salts, along with almost unchanged NaCl rejection, is characteristic of dielectric exclusion.

A notable effect of the PEI coating was also observed regarding pure water permeance (PWP). Compared to the baseline TFC membrane (2.0 LMH/bar), the PWP decreased to 1.25 LMH/bar with the PEI-600

coating, consistent with the formation of a denser selective layer due to internal charge compensation, but also a consequence of increased film thickness (cf. Section 3.5). Interestingly, the PEI-25k membrane exhibited an increased PWP of 2.7 LMH/bar. This suggests that the high molecular weight PEI subtly alters the film structure in a way that facilitates water transport (cf. Section 3.5). These distinct behaviors underscore the importance of polyelectrolyte molecular weight and interaction mode in tailoring membrane performance. Overall, at 1 g/L solute concentration, the salt rejection observed for the PEC NF membranes was higher than that for the TFC membranes for all salts, demonstrating the value of the PEC approach in increasing solute rejection and modulating ion-selectivity.

Crucially, the long-term chemical stability of these membranes under active chlorine cleaning/disinfection conditions was also assessed (Fig. 6). Nexar TFC and PEC membranes were subjected to continuous filtration with a 500 ppm NaOCl solution (containing 2 g/L NaCl, pH 9) for up to 2 days. No significant decline in membrane performance, regarding flux or salt rejection, was observed for any of the membranes. The Nexar TFC membranes retained salt rejection and PWP within experimental error. The PEC membranes interestingly revealed an increase in permeance by about 20 % while rejection remained unchanged. Overall, this indicates a high resistance to hypochlorite attack. This stability is attributed to the chemical composition of the selective films: Nexar is a hydrocarbon polymer containing sulfonic acid groups, and PEI is a polyamine; neither polymer contains the aromatic amide bonds that are the Achilles' heel of polyamide membranes. For example, van Do et al. reported that even 1 ppm of chlorine could cut the life of a polyamide NF membrane significantly [7]. Prior studies have also shown that sulfonated poly(arylene) membranes and polyelectrolyte multilayer membranes can withstand active chlorine much better than polyamides [15,16,35]. The findings of this study emphasize that using an ionomer-based PEC membrane is a viable strategy for applications requiring active chlorine usage.

3.5. Polyelectrolyte complex membranes – structure characterization

To investigate how the structural and interfacial properties of the membranes relate to their separation performance, a comprehensive set of physical characterizations was performed on TFC, PEC PEI-600, and PEC PEI-25k membranes. First of all, SEM images of the membrane cross-sections revealed that compared to the thickness of the Nexar film in the TFC membrane (1.45 ± 0.10 μm ; cf. Table 3), such film thickness values were 1.44 ± 0.15 μm for PEC PEI-25k and 1.70 ± 0.13 μm for PEI-600 (cf. section S2 in SI). Hence, the PEI with high molecular weight did not lead to a significant change while the PEI with low molecular weight caused an increase of film thickness by about 15 %. This indicates that a significant fraction of PEI-600 has been taken up by the Nexar film, likely because the nanochannels are accessible for the small PEI

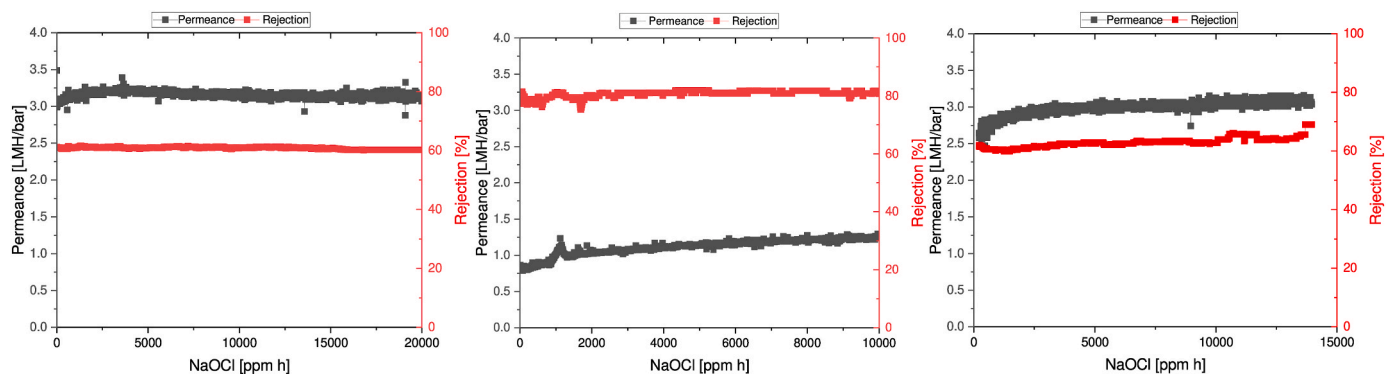


Fig. 6. Permeance and overall salt rejection (based on conductivity measurements) during nanofiltration of a solution containing 0.5 g/L NaOCl and 2 g/L NaCl at pH 9. Left Nexar TFC, Middle PEC PEI-600, Right PEC PEI-25k

molecules and their diffusivity is high [36]. The resulting increase of film thickness will contribute to the observed reduction in PWP (cf. Fig. 5). In contrast, PEI-25k may have just bound in the surface region of the Nexar film.

Furthermore, zeta potential measurements were conducted as a function of pH to assess surface charge (Fig. 7), water contact angles were measured to quantify surface wettability, depth profiling by TOF-SIMS and XPS was used to examine the distribution of PEI within the selective layer, and AFM imaging (in the wet state) was carried out to visualize membrane surface morphology (Fig. 8). These complementary techniques were selected to elucidate the effects of PEC formation onto membrane structure, and to distinguish between the film-embedded and surface-localized PEI configurations evoked above.

The Nexar TFC membrane shows a negative zeta potential across the pH range 3–10. At low pH (~3), the zeta potential is slightly less negative (around -30 mV) because some sulfonic acid groups may be protonated. As pH increases, the sulfonic acid groups deprotonate fully and impart a strong negative charge (zeta potential reaching about -100 mV at pH 10). There is no isoelectric point observed in this range; the curve is typical for a strong cation-exchange surface. This confirms that the Nexar film in the TFC membranes is anionic (negatively charged). To ensure that zeta potential data are reproducible, second batches of all membranes had been analyzed (see Fig. S7, Section S3 in SI), and the results confirm that the differences between the three membranes are significant.

The PEC PEI-600 membrane, interestingly, does not show a positive zeta potential despite the presence of PEI, a weak polycation, with amino groups that are protonated (positively charged) in the lower pH range. Its zeta potential curve is even slightly more in the negative range than that of the TFC membrane. This difference is considered significant because the same was observed for a second batch of TFC and subsequently prepared PEC membrane (cf. Fig. S7). While this appears counterintuitive, one plausible explanation is that the low-molecular-weight PEI diffuses into the Nexar matrix and binds to internal sulfonic acid groups, forming a homogeneous PEC throughout the film. In fact, the conditions used in the PEI dip-coating step, low concentration and short time, have very likely not yielded a stoichiometric compensation of all Nexar charges, and a subsequent redistribution of the PEI in the film has resulted in a net-negative structure. The low-MW PEI certainly does not form a distinct topcoat on the Nexar film; that the surface still exposes much more sulfonic acid groups may have been caused by a reorganization of the film as consequence of PEC formation.

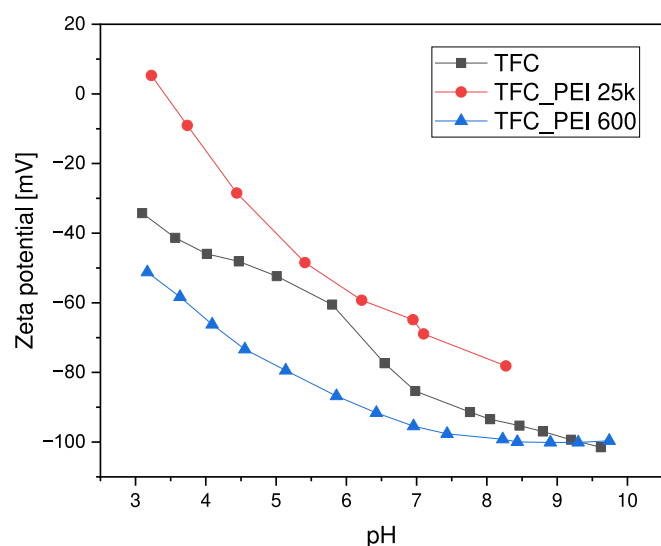


Fig. 7. Zeta potential as function of pH value for Nexar TFC, PEC PEI-600 and PEC PEI-25k membranes (data for first batch; all data in Fig. S7).

The PEI-25k membrane, on the other hand, exhibits a significantly more positive zeta potential compared to the Nexar TFC membrane at all pH values (Fig. 7). However, for a second batch of TFC and subsequently prepared PEC membrane (cf. Fig. S7), zeta potential was more positive only at pH < 4. A plausible interpretation is that the high-MW PEI (25 kDa) is forming a thin layer containing both PEI amino groups and Nexar sulfonic acid groups at the outer surface. Apparently, PEI-25k does not fully neutralize all sulfonic acid groups. Analogous to what has been discussed above for PEI 600, redistribution of PEI has likely resulted in a net-negative surface structure.

Water contact angle data provide insight into surface hydrophilicity; the following values were measured: TFC $89.6 \pm 3.4^\circ$, PEI-600 $78.2 \pm 4.5^\circ$, PEI-25k $43.1 \pm 7.4^\circ$. The TFC membrane's contact angle of $\sim 90^\circ$ indicates a relatively hydrophobic surface (consistent with the hydrophobic matrix consisting of polystyrene and polybutadiene blocks dominating the wetting properties of the surface). The PEC PEI-600 membrane shows a moderate reduction in contact angle, suggesting that the surface became somewhat more hydrophilic by integration of PEI-600 in the PEC film. That the difference compared to Nexar is only small aligns with the idea that PEI-600 is mostly below the outermost surface, it increased the overall hydrophilicity of the film, but the surface is still dominated by the polymer segments of the Nexar matrix. The PEC PEI-25k membrane, however, has a much lower contact angle ($\sim 43^\circ$), indicating a hydrophilic surface. This would be in line with the presence of many hydrophilic functional groups (primary, secondary and tertiary amino groups as part of the branched PEI structure, and also Nexar sulfonic acid groups; cf. Fig. 7) at the surface, possibly yielding a water-swollen layer on top of the Nexar-based film.

TOF-SIMS along with argon ion sputtering was employed to investigate the spatial distribution of PEI within the PEC membranes; the Nexar TFC membrane was studied as reference (see Fig. S9, section S5 in SI). The CN^- fragment served as a marker for nitrogen-containing species, particularly PEI. First, for all membranes after a constant value for a certain sputtering period, a large increase of CN^- intensity was observed that corresponds to reaching the polyacrylonitrile (aPAN) support membrane surface (with a much higher CN^- yield compared to the Nexar-PEI complex). Notably, the onset of that increase occurred at similar same sputter time (~ 1000 s vs. 1200 s) for both the TFC reference and the PEI-25k membrane, which is consistent with their similar film thickness according to SEM, including the minor batch-to-batch variations originating from the dip-coating process (cf. above). In contrast, the PEI-600 membrane showed a delayed PAN signal onset (~ 2000 s), indicating a significantly thicker Nexar-PEI film, in line with the SEM data (cf. above). Second, the PEI-25k-modified membrane showed a strong CN^- signal only in the initial ~ 50 s of sputtering, followed by a sharp decrease to baseline, i.e., the intensity observed for Nexar that does not contain any nitrogen (and CN^-). This indicates that PEI-25k is confined to the near-surface region without penetrating deeper into the Nexar film. An estimation based on sputter times (50 s for PEI-containing part and 1000 s for entire film) and film thickness from SEM ($1.4 \mu\text{m}$) yields a thickness of the PEC complex layer of max. 70 nm . In contrast, for the PEI-600-coated membrane, the CN^- signal was significantly higher compared to the data for PEI-25k and Nexar and remained stable throughout the sputtering until reaching the PAN support, indicating a uniform integration of the low-molecular-weight PEI throughout the Nexar film. Hence, the TOF-SIMS data clearly demonstrate that two different PEC architectures, i.e., film-embedded versus surface-localized PEI configurations, have been obtained.

XPS analysis along with argon ion sputtering was also employed to investigate the spatial distribution of PEI within the PEC membranes; the Nexar TFC membrane was studied as reference (see Fig. S10, section S6 in SI). First, the TFC membrane (only Nexar on top surface) was analyzed and then sputtered for a short time of $\sim 15 \text{ min}$. As expected for the copolymer (cf. Fig. 1), besides carbon, small fractions of oxygen (between 2 and 4 %) and sulfur ($\sim 2 \%$) were found, the content of nitrogen was below the detection limit. For the PEC PEI-600 membrane no

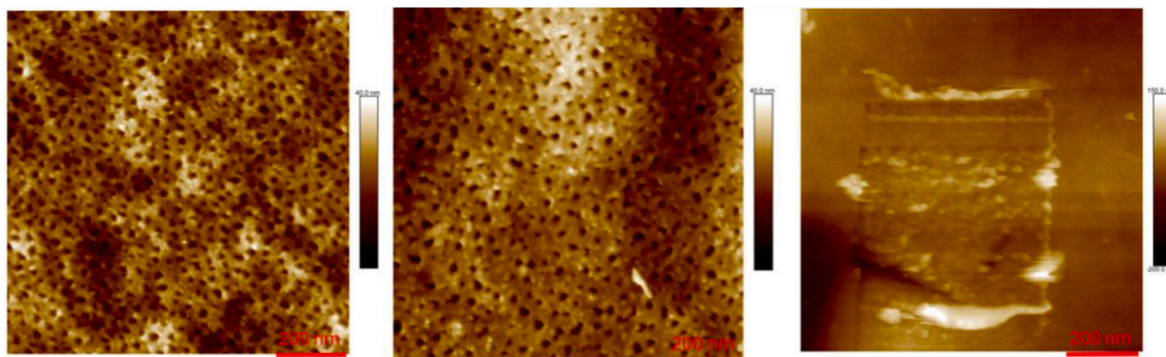


Fig. 8. AFM analysis of membranes in wet state. Left TFC, Middle PEC PEI-600, Right PEC PEI-25k

significant nitrogen content was found at the top surface (sputter time 0 min). The intensity of the nitrogen signal sharply increased after start of sputtering and then fluctuated between lower than detection limit and $\sim 2\%$ before reaching a stable value of about 0.5% until a sputter time of 120 min. Because exactly the same fluctuation between higher and lower values was observed for N, S and O, the distribution of PEI is most likely aligned with the distribution of sulfonic acid groups as it would be expected for PEC formation. Notably, for the PEC PEI-25k membrane a nitrogen content beyond the detection threshold was only detected

during the first ~ 15 min of sputtering, indicating a top layer of PEI on the surface of the Nexar film. Overall, the XPS results confirm the conclusions from the TOF SIMS analyses that PEI 600 is distributed in the Nexar film while PEI 25k is confined on the film surface.

AFM was used to image the surfaces of the membranes under hydrated conditions (Fig. 8). The TFC and PEC PEI-600 membranes showed a distinct nanostructure, with domains on the order of ~ 10 – 25 nm, likely corresponding to ionic clusters/hydrophilic domains within the hydrophobic Nexar matrix. Quantitative analysis of the AFM images

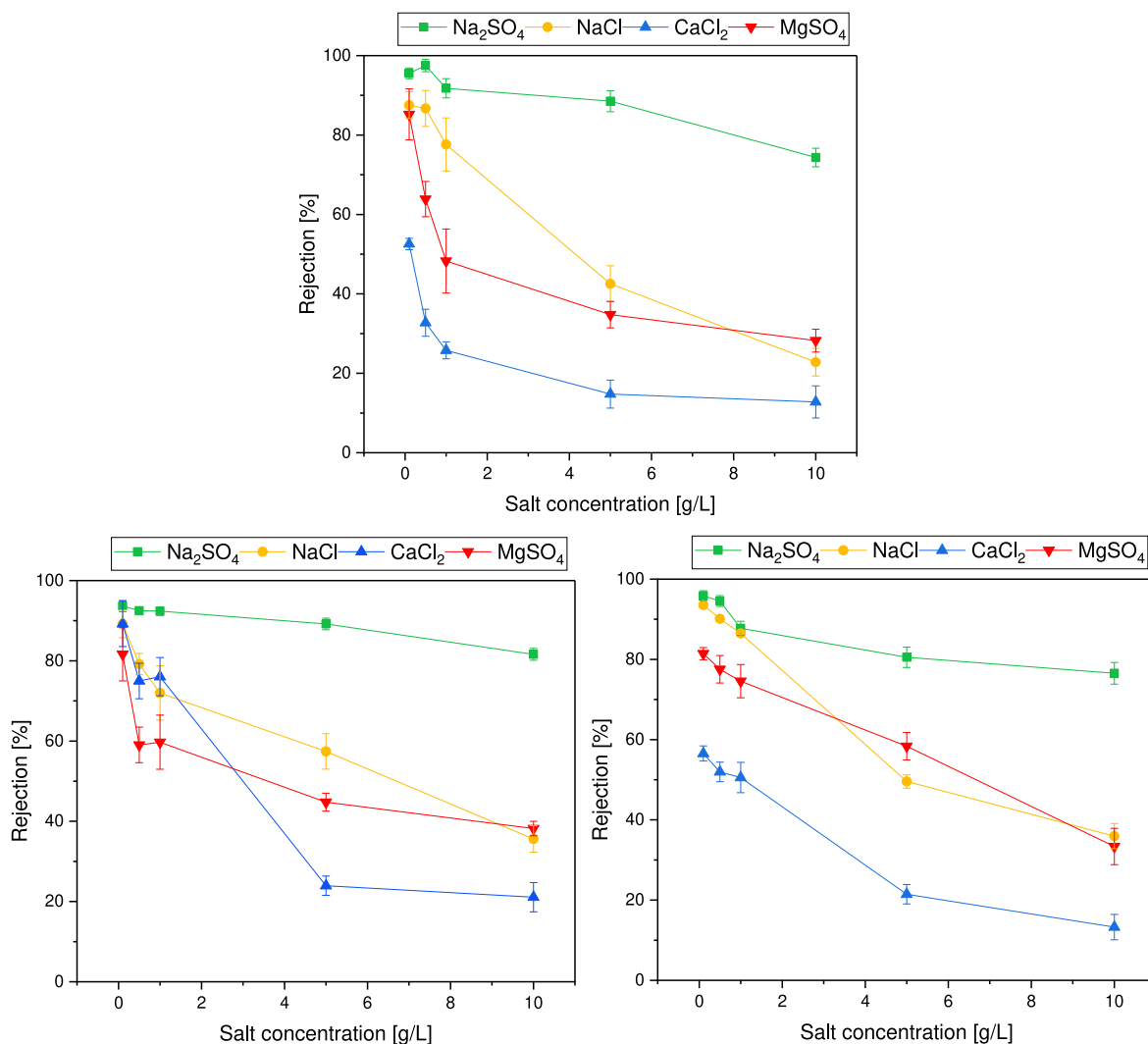


Fig. 9. Salt rejection as a function of salt concentration for Nexar TFC (top), PEC PEI-600 (bottom-left), and PEC PEI-25k membranes (bottom-right).

using ImageJ revealed a size of 17.4 ± 0.6 nm and an surface area coverage of 16.8 ± 1.6 % for the TFC membrane. The PEI-600 membrane exhibited a similar surface morphology, though slightly smoother and with fewer domains per area. ImageJ analysis indicated an increased average size of 20.3 ± 1.2 nm and a larger surface area percentage of 24.8 ± 3.0 %. This suggests that PEI-600 incorporation into the Nexar film does not much change the kind of nano-scale segregated morphology. Apparently embedding significant fractions of PEI in the film, likely in the hydrophilic domains/channels, requires some rearrangement, but the Nexar still defines the matrix. This is analogous to structural changes deduced from SAXS analyses of Nafion/polyallylamine PEC films compared to the pristine Nafion precursor [36]. The PEI-25k membrane proved difficult to image reproducibly by AFM. Under wet conditions, high-force tapping caused displacement of material on the surface, suggesting that the PEI-25k layer is very soft. This observation supports the interpretation that PEI-25k that is anchored in a thin PEC layer in the Nexar film forms a thin water-swollen coating that is not fully interpenetrated into the solid Nexar matrix.

3.6. Influence of feed and filtration conditions on separation performance

To gain deeper insight into the solute rejection behavior, experiments were conducted covering a range of feed concentrations (from 0.2 up to 10 g/L) for various salts using the three membrane types. Fig. 9 shows the rejection as a function of salt concentration for NaCl, Na_2SO_4 , CaCl_2 , and MgSO_4 for Nexar TFC vs PEC PEI-600 and PEC PEI-25k membranes. At low salt concentration (~ 0.2 – 1 g/L), all membranes exhibit the highest rejections. The net negatively charged NF membranes (cf. Fig. 7) operate predominantly under Donnan exclusion in this regime: the fixed charges in the membrane repel co-ions (anions) and attract counter-ions, leading to high rejection of salts, especially those with multivalent co-ions [37]. As the feed salt concentration increases, a decline in rejection is observed for all salts and all membranes. For example, NaCl rejection for TFC drops from ~ 85 % at 0.2 g/L to ~ 50 % at 5 g/L and ~ 20 % at 10 g/L. The PEC membranes similarly show decreasing rejection, though they maintain higher absolute rejection than TFC at each concentration. Importantly, the changes are completely reversible, as confirmed by identical NaCl rejection at 1 g/L feed concentration after filtrations had been performed at concentrations up to 10 g/L (see section S7 in SI). While this behavior is often interpreted in terms of electrostatic screening of fixed charges and thereby reducing the magnitude of the Donnan effect, this explanation is not entirely applicable to ionomer- or PEC-based membranes. Instead, the dominant mechanism appears to be increased salt partitioning into the membrane, as intrinsic charge compensation, i.e., pairing of fixed charged groups of Nexar and PEI, is gradually overtaken by extrinsic compensation by mobile ions of the salt from the external solution, an effect commonly referred to as the „anti-polyelectrolyte“ effect. This phenomenon leads to greater salt uptake and internal swelling, which in turn reduces salt rejection and enhances permeance [12].

Indeed, the rejections of TFC, PEI-600, and PEI-25k membranes start to converge at the highest concentrations, indicating that the advantage of forming the PEC with PEI diminishes when charge screening is strong. Nonetheless, even at 10 g/L, the PEC membranes still show modestly higher rejection than TFC ones (for MgSO_4 , ~ 40 % vs 30 %; for NaCl, ~ 35 % vs 20 %). This suggests the PEC membranes have a more robust exclusion capacity across a range of ionic conditions, likely due to their combination of charge and smaller pore size. Looking at differences among the salts: Na_2SO_4 consistently has the highest rejection on all membranes at all concentrations, owing to the divalent sulfate anion being strongly excluded by Donnan effects and also contributions of size exclusion and dielectric exclusion (sulfate has a larger hydrated radius than Cl^-). CaCl_2 is less rejected because the cation is multivalent. MgSO_4 and NaCl show intermediate rejections, with NaCl usually above MgSO_4 . These detailed ion-specific effects are complex, but generally follow known NF trends [38,39]. To summarize, the salt rejection results

indicate that the membranes operate under a Donnan exclusion dominated regime at low concentration. At higher salt concentration, rejection is increasingly governed by salt uptake and swelling within the membrane. The conversion of the Nexar to a PEI-PEC film shifts the performance to higher rejection for all solutes, acting like a membrane with a more neutral charge pattern and smaller pores, that also increase the effect of dielectric exclusion. The observations in this study agree with the trends reported by Zelner et al. for a Nafion-based mixed-charge PEC NF membrane: high rejections at low salt concentrations and convergence of rejections at high ionic strength, with multivalent ions always more rejected than monovalent [12,40].

Permeate fluxes were found to increase with salt concentration for all membranes. The increasing osmotic pressure at constant applied pressure was considered to calculate the solution permance at the actual transmembrane pressure as driving force; data are shown in Fig. 10. Importantly, the changes are completely reversible, as confirmed by identical salt solution permeance at 1 g/L feed concentration after filtrations had been performed at concentrations up to 10 g/L (see section S7 in SI). While the Donnan model is useful for interpreting ion rejection through electrostatic effects, it does not explain the observed increase in solution permeance with increasing salt concentrations. Nexar contains sulfonic acid group-containing segments (cf. Fig. 1) that cluster in the hydrophilic domains that function as selective pathways through the membrane (cf. Section 3.2). PEC formation modulates these domains toward a mixed-charge character. The increased salt partitioning and connected water uptake leads to swelling, resulting in increased free space within the membrane's charged domains, reducing resistance to pressure-driven water transport. Indeed, the extent of permeance increase varied with salt type, reflecting differences in ion valence and hydration, and resulting interactions with the membrane. NaCl exhibited the most pronounced permeance increase. In contrast, multivalent salts such as Na_2SO_4 , CaCl_2 , and MgSO_4 caused only moderate increases in permeance. This cannot directly be related to the extent of their partitioning into the membrane because, e.g., at the same high molar salt concentration the passage of NaCl was lower than that of CaCl_2 , i.e., the salt concentration in the membrane was lower for NaCl. Membrane structure also significantly influenced how permeance responded to salt concentration. The TFC membrane displayed the steepest increase in permeance for NaCl, consistent with its relatively open pore structure and high sensitivity to ionic screening of the charges of sulfonic acid groups in the channels. The PEI-600 membrane showed a more uniform and lower response across all salts. This may be due to its tighter morphology and the presence of interpenetrated PEI, which associates with sulfonated Nexar segments and thereby reduces salt-specific effects. The PEI-25k membrane exhibited a response similar to the TFC membrane, with a strong NaCl-induced permeance increase and moderate effects from multivalent salts. This suggests that while the PEI-25k surface layer modifies charge interactions, it does not significantly modulate the transport properties of the underlying Nexar layer.

In order to further clarify the influence of the barrier film on solute rejection, nanofiltration experiments were conducted using sucrose solutions at varying NaCl concentrations (see Fig. 11). Importantly, sucrose rejection remained constant across all NaCl concentrations, maintaining the same value observed for sucrose-only tests for each membrane type (TFC, PEI-600, and PEI-25k). In contrast, NaCl rejection decreased and solution permeance increased with increasing NaCl concentration (see section S8 in SI), consistent with the single-solute experiments (cf. Figs. 9&10). These results are noteworthy. They confirm that the size-selective component of the barrier film remains functionally stable, even under conditions where salt passage and water permeance increase significantly. This indicates that polymer relaxation and channel reorganization in response to salt uptake and resulting local swelling do not significantly alter the size of the limiting apertures within the membrane. Instead, the constant rejection of sucrose suggests that additional transport pathways with similar size-exclusion properties may open upon swelling. This behavior likely reflects the nano-scale

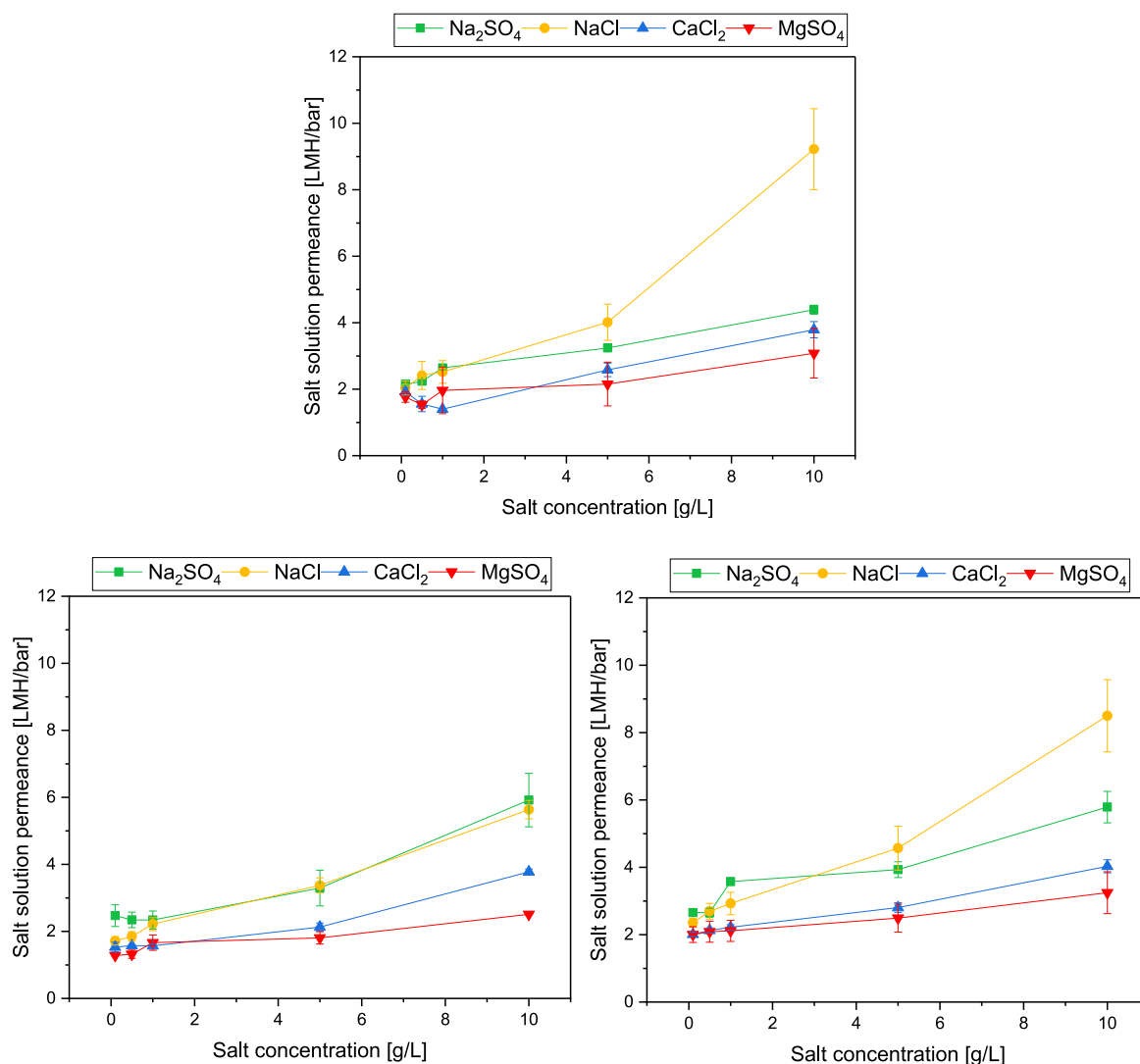


Fig. 10. Calculated permeance of different salt solutions as a function of salt concentration for Nexar TFC (top), PEC PEI-600 (bottom-left), and PEC PEI-25k membranes (bottom-right).

morphology that has been formed under the Nexar casting conditions, consisting of spherical sulfonated domains embedded within a continuous matrix of the hydrophobic blocks [26], connected by apertures that from an array of size-selective “bottlenecks”. As a result, the membrane can exhibit significant increases in permeance without compromising its size-selective function, demonstrating that water transport and size selective solute rejection are partially decoupled. This offers interesting perspectives because the productivity (proportional to permeance) of nanofiltration is higher at higher salt concentration without compromising size selectivity, and a higher selectivity between neutral and charged solutes is expected for higher salinity of the feed.

3.7. Organic micropollutant rejection

Another critical application target for the PEC-NF membranes is the removal of organic micropollutants (OMP) from water. Six model OMP were tested, selected to represent a range of sizes, hydrophilic/phobic balances, and charged states (see Fig. 12; for chemical structure see section S9 in SI). Molecular weight (MW) ranged from 206 to 290 Da, placing all compounds within the range where nanofiltration can typically enable only partial rejection. Rejection data for the three membrane types are shown in Fig. 12. First of all, it is obvious that the PEC membranes outperform the Nexar TFC membrane in all cases. For

example, carbamazepine (MW 236 Da) was rejected at ~25 % by the TFC membrane, compared to ~50 % and ~35 % by the PEI-600 and PEI-25k membranes, respectively. Triclosan (MW 290 Da) exhibited the highest rejection among the six compounds, with values of ~60 % for TFC, ~60 % for PEI-600, and ~80 % for PEI-25k. For ibuprofen, despite having the smallest MW (206 Da), higher rejection than for four other OMP was observed, particularly for the PEC membranes (~65 % for PEI-25k). These results reflect the combined effects of three mechanisms.

Size exclusion. OMP rejection generally increased with molecular weight. Larger substances such as triclosan were more effectively retained than smaller ones like ibuprofen or carbamazepine. This trend aligns with the barrier film pore sizes inferred from neutral solute rejection. The PEC membranes showed higher rejection for sucrose (cf. Section 3.2) and likewise demonstrated higher rejection for OMP. However, for all OMP except carbamazepine, the PEI-25k membrane showed higher rejection than the PEI-600 membrane, in contrast to the results for sucrose (cf. Fig. 5). This indicates that other effects have also an impact.

Charge effects. At pH 7, two of the OMP, i.e., sulfamethoxazole and ibuprofen, carry a negative charge. As all membranes exhibit net negative surface charge, electrostatic repulsion is expected to contribute to their rejection. The measured zeta potentials (cf. Fig. 7) show that the TFC membrane is strongly negative, PEI-600 is slightly more negative,

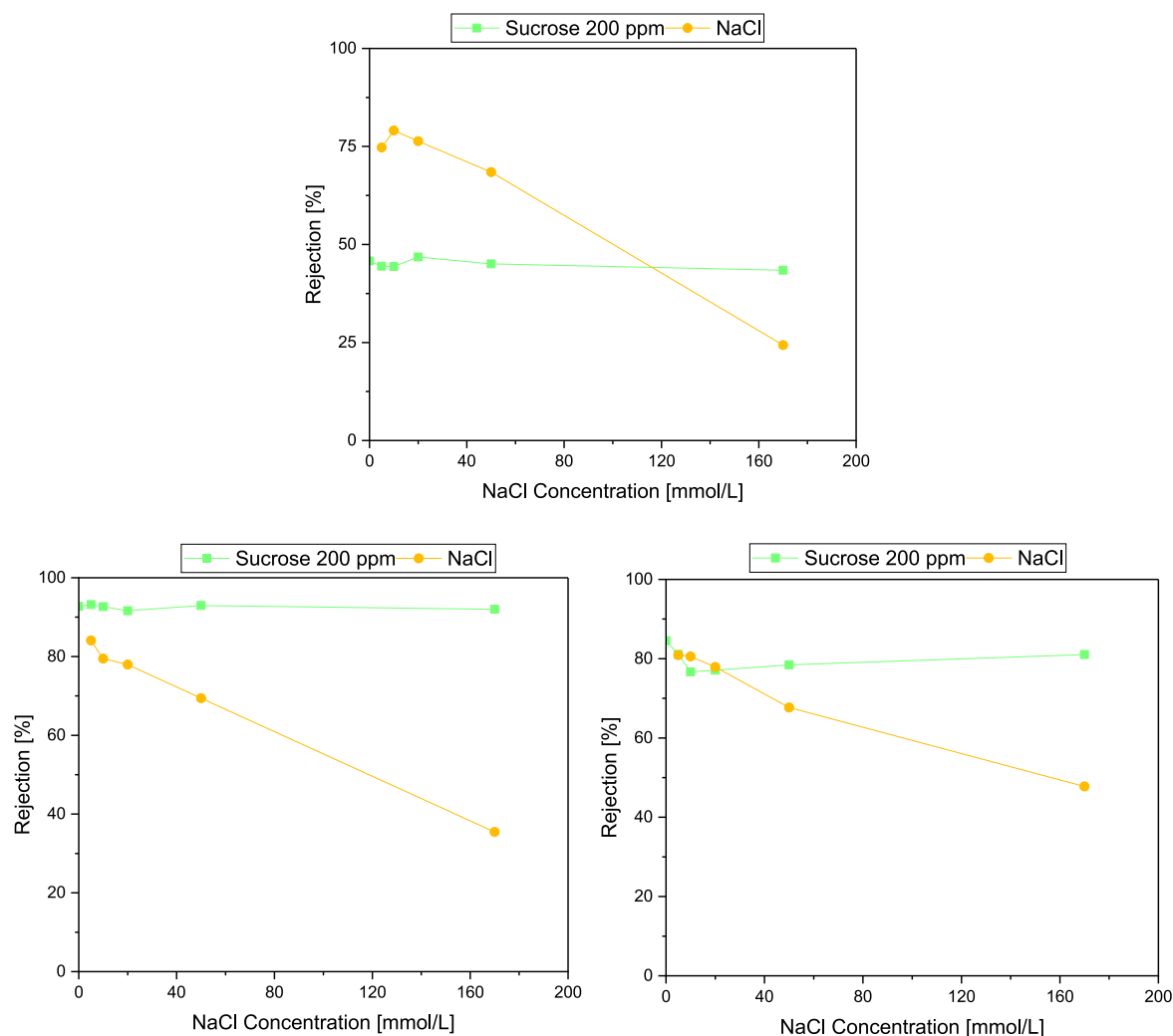
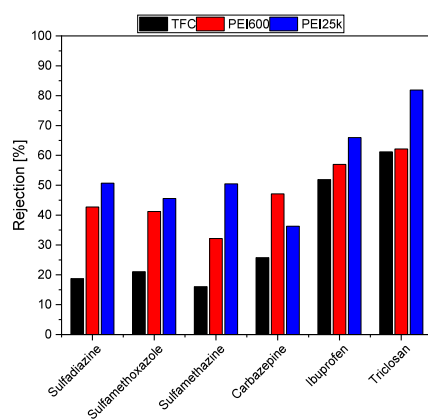


Fig. 11. Rejection for sucrose (0.2 g/L) and NaCl (concentration varied from 0.2 to 10 g/L) used as mixture in nanofiltration through Nexar TFC (top); PEC PEI-600 (bottom-left) and PEC PEI-25k (bottom-right) membranes.



	Sulfadiazine	Sulfamethoxazole	Sulfamethazine	Carbamazepine	Ibuprofen	Triclosan
log K _{ow}	0.21	0.86	1.62	2.45	3.5	4.86
MW [Da]	250.3	253.3	278.3	236.3	206.3	289.5
net charge at pH 7	Neutral	negative	neutral	neutral	negative	neutral

Fig. 12. Rejection of OMP of different size, net charge and hydrophilicity/phobicity for Nexar TFC, PEC PEI-600 and PEC PEI-25k membranes. Cross flow filtration at 15 bar with either the mixture of the first four or that of ibuprofen and triclosan, all at pH 7.

and PEI-25k is notably less negative (i.e., closer to neutral). Nevertheless, the PEI-25k membrane achieved higher rejection of anionic OMP, indicating that electrostatic repulsion has not the decisive impact.

Hydrophobic interactions and partitioning into the membrane. Triclosan and ibuprofen, with log K_{ow} values of 4.86 and 3.5, respectively, are the most hydrophobic substances in the set. For these two OMP, rejection was much higher than for the other four OMP with the Nexar TFC membrane; this major difference does neither fit to the size nor to the charge differences. Instead, it is plausible that the partitioning of the more hydrophobic solutes into the charged/hydrophilic transport channels of the Nexar film is less preferred than that of the more hydrophilic solutes. The increase in rejection for the PEC compared to the Nexar TFC membrane can be related to the decreasing pore size. However, the markedly higher rejection for the most hydrophobic solutes (triclosan and ibuprofen) for the PEC PEI-25k compared to the PEC PEI-600 membrane does not fit to their effective pore sizes (larger for PEC PEI-25k). This additional impact on rejection can be related to the much higher hydrophilicity of the PEC PEI-25k compared to the PEC PEI-600 surface (water contact angle 43° vs. 78°; cf. Section 3.5), that is explained by the PEI layer on top of the barrier film. These findings are in line with prior work by Huang et al., who demonstrated that hydrophilic surface modification using polydopamine improved rejection of neutral, hydrophobic OMP by NF membranes due to suppression of attractive membrane-solute interactions [41]. In that study, transmission of hydrophobic OMP was reduced by up to 70 % after hydrophilic modification, comparable to the improvements observed here for the PEC compared to the TFC membranes.

4. Conclusions

This study developed a new type of polyelectrolyte complex thin-film composite nanofiltration (PEC-NF) membranes using a scalable two-step dip-coating process. A porous aminated PAN support (aPAN) was first coated with the anionic ionomer Nexar to form a nanostructured TFC membrane. Under optimized toluene-rich solvent conditions and controlled drying, the Nexar film formed a continuous anionic/hydrophilic channel-in-matrix structure, yielding a membrane with a moderate MWCO (~450 Da) and salt rejection dominated by Donnan exclusion. Adding a cationic polyelectrolyte, i.e., branched polyethylenimine (PEI), tightened the membrane barrier structure and introduced mixed-charge functionality. The resulting PEC NF membranes exhibited substantially lower MWCO (~350 Da with PEI-600) and higher rejection of salts (particularly those with divalent ions) as well as organic micropollutants. Thereby, this study also demonstrates that the concept of ionomer/polyelectrolyte PEC-NF membranes that had been established with the fluorinated ionomer Nafion can be extended to a more sustainable fluorine-free alternative. Increasing feed salt concentration lead to pronounced reduction of salt rejection and increase of solution permeance, but, remarkably, size selectivity (judged by rejection of the neutral solute sucrose) remained unchanged. All fabricated membranes showed excellent chemical durability, with no loss in flux or rejection after exposure to 500 ppm NaOCl (pH 9) for 48 h. The combined findings of this work have important implications for the design of next-generation NF membranes. First, employing a hydrocarbon ionomer (e.g., Nexar) as the base selective film provides a robust, chlorine-tolerant alternative to traditional polyamide films. Second, the formation of a PEC with a polyelectrolyte (e.g., PEI) can be utilized to significantly alter the solute rejection pattern without the need for multi-step layer-by-layer assembly. By selecting polyelectrolyte's molecular weight and deposition conditions, one can adjust the degrees of interpenetration in the film versus surface coverage of the ionomer, thereby controlling pore size and charge of the selective film. This tunability enables different extents of improvement in salt and micropollutant rejection: The PEC-NF membranes showed changed multivalent salt rejection due to modulated Donnan exclusion, as well as improved OMP rejection thanks to a smaller effective pore size. PEC-NF membranes

with a more hydrophilic surface show substantially higher rejection of neutral hydrophobic contaminants. Overall, the two-step fabrication of PEC membranes from complementary ionomer and polyelectrolyte is a viable strategy to create NF membranes that combine tunable selectivity and permeance with superior chemical resistance for demanding water reuse applications.

CRediT authorship contribution statement

Sven Kroß: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Viatcheslav Freger:** Writing – review & editing, Conceptualization. **Mathias Ulbricht:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.memsci.2025.124698>.

Data availability

Data will be made available on request.

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